TESTING AN ETALE LOCAL ON THE SOURCE-AND-TARGET PROPERTY OF MORPHISMS ON ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is a meta-statement about an arbitrary property P; the proof handles each of the three equivalence claims in its own paragraph, and the closing paragraph on representable f is part of the same proof environment.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etale localness on the source-and-target characterized through a double diagram. Descent, Lemma descent-lemma-local-source-target-characterize, used at proof lines 4372-4376 and 4379-4382. Is the engine of both the (2) implies (1) direction and the representable comparison.
- Lifting a morphism from an algebraic space to a presentation by a scheme. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 4343-4345. Makes (1) implies (2) immediate by exhibiting a diagram with U -> X surjective.
- The definition of a property of representable morphisms in the functorial sense. Section section-representable, used at proof lines 4336-4339. Fixes the third equivalent formulation in the statement.
- Base change and fppf locality of the property. used at proof lines 4385-4390. Allows the final comparison for representable f.

Source reference. Original lemma 04R1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6169–6169. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $\mathcal{P}$ be a property of morphisms of schemes which is étale local on the source-and-target. Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Consider commutative diagrams*

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale. The following are equivalent

1. *for any diagram as above the morphism h has property $\mathcal{P}$, and*
2. *for some diagram as above with $a : U \to X$ surjective the morphism h has property $\mathcal{P}$.*

If X and Y are representable, then this is also equivalent to f (as a morphism of schemes) having property $\mathcal{P}$. If $\mathcal{P}$ is also preserved under any base change, and fppf local on the base, then for representable morphisms f this is also equivalent to f having property $\mathcal{P}$ in the sense of Section 03HA.

Proof. Let us prove the equivalence of (1) and (2). The implication (1) $\Rightarrow$ (2) is immediate (taking into account Spaces, Lemma 02X1). Assume

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array} \qquad \begin{array}{ccc} U' & \xrightarrow[h']{} & V' \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

are two diagrams as in the lemma. Assume $U \to X$ is surjective and h has property $\mathcal{P}$. To show that (2) implies (1) we have to prove that h' has $\mathcal{P}$. To do this consider the diagram

$$\begin{array}{ccccc} U & \longleftarrow & U \times_X U' & \longrightarrow & U' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle (h,h')} & & \downarrow{\scriptstyle h'} \\ V & \longleftarrow & V \times_Y V' & \longrightarrow & V' \end{array}$$

By Descent, Lemma 04R1 we see that h has $\mathcal{P}$ implies (h, h') has $\mathcal{P}$ and since $U \times_X U' \to U'$ is surjective this implies (by the same lemma) that h' has $\mathcal{P}$.

If X and Y are representable, then Descent, Lemma 04R1 applies which shows that (1) and (2) are equivalent to f having $\mathcal{P}$.

Finally, suppose f is representable, and U, V, a, b, h are as in part (2) of the lemma, and that $\mathcal{P}$ is preserved under arbitrary base change. We have to show that for any scheme Z and morphism $Z \to X$ the base change $Z \times_Y X \to Z$ has property $\mathcal{P}$. Consider the diagram

$$\begin{array}{ccc} Z \times_Y U & \longrightarrow & Z \times_Y V \\ \downarrow & & \downarrow \\ Z \times_Y X & \longrightarrow & Z \end{array}$$

Note that the top horizontal arrow is a base change of h and hence has property $\mathcal{P}$. The left vertical arrow is étale and surjective and the right vertical arrow is étale. Thus Descent, Lemma 04R1 once again kicks in and shows that $Z \times_Y X \to Z$ has property $\mathcal{P}$. $\square$